

Integration and development of AI Accelerator applications in Universal Robots arm.

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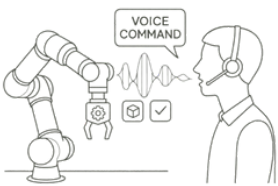
Artificial Intelligence
Final Degree Project

Introduction

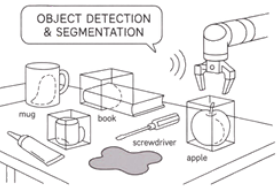
This work presents an **autonomous robotic system** to enhance the autonomy of individuals with reduced mobility through **voice-controlled object manipulation**. A Universal Robots UR10e arm is linked to an NVIDIA Jetson AI Accelerator that runs an end-to-end system of AI models, so that a spoken command ("Pass me the ball") is translated into a safe and precise picking of the object. The system uses **Whisper** to transcribe user's speech, **Qwen3**, a small language model extracts the intent, a **YOLOv8** detects the objects and with **Segment Anything (SAM)** objects masks are obtained and the ideal grasp point is computed transforming this 2D coordinate into 3D with the hand-eye camera calibration.



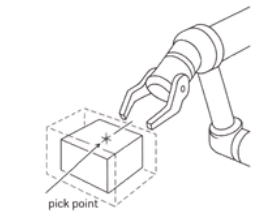
Objectives



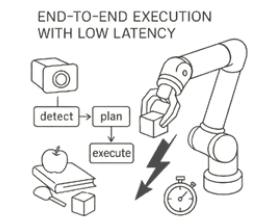
Voice-commanded control of UR arm.



Accurate object detection and segmentation in cluttered scenes.

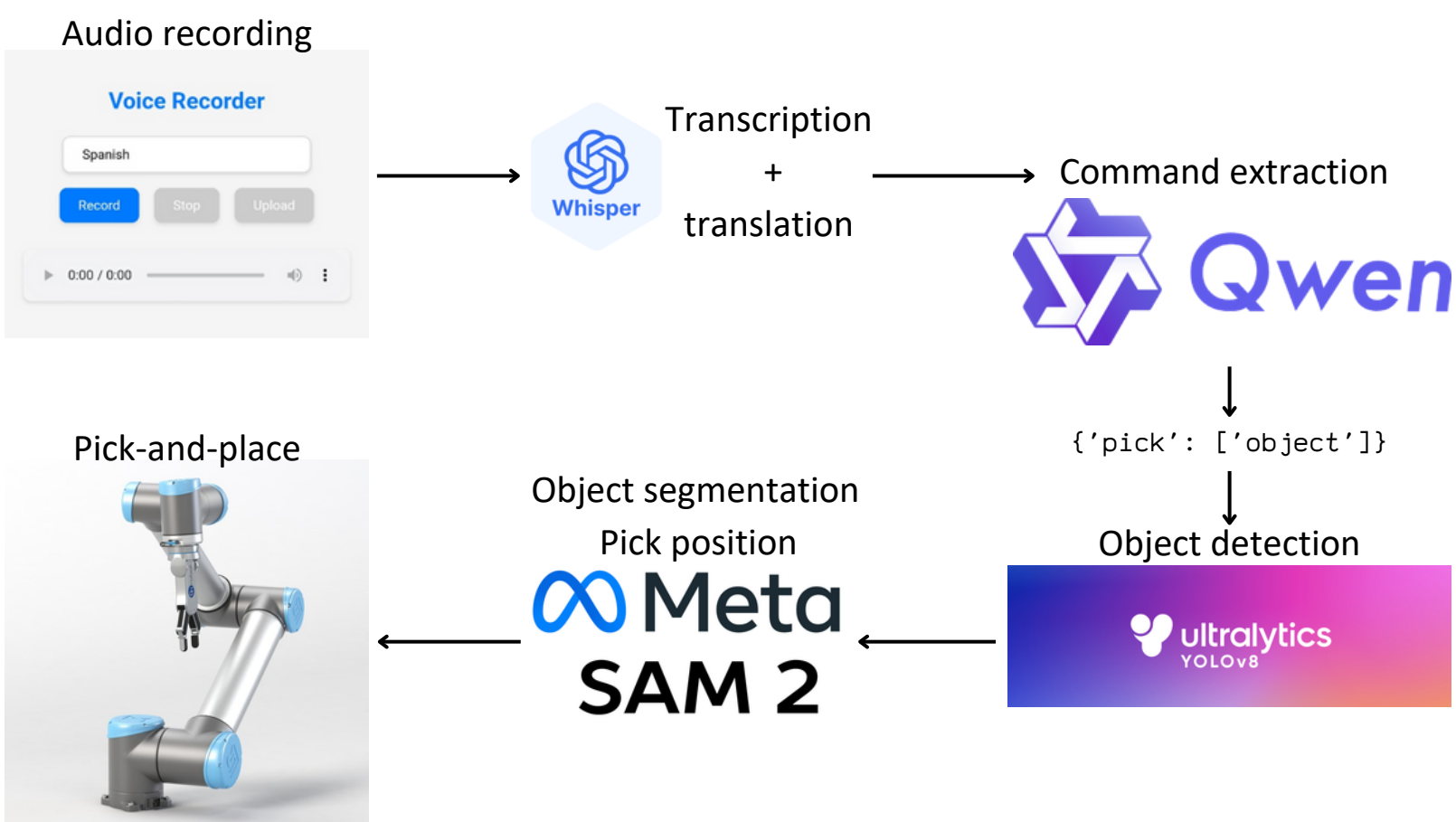


Object grasp planner for pick point.



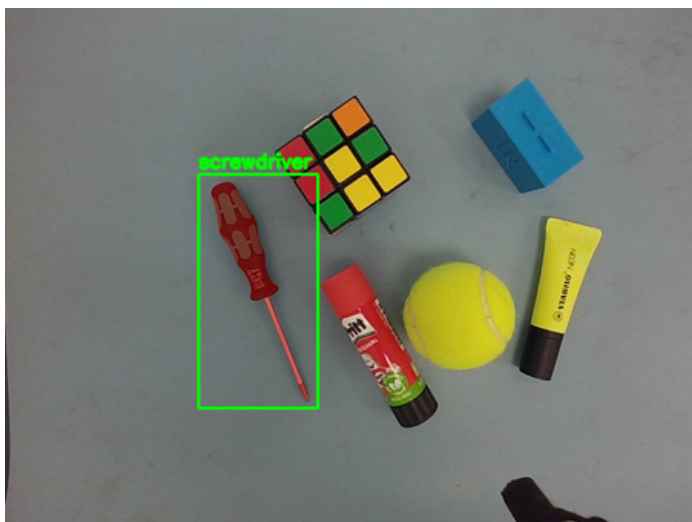
End-to-end execution with low latency.

Developed Pipeline

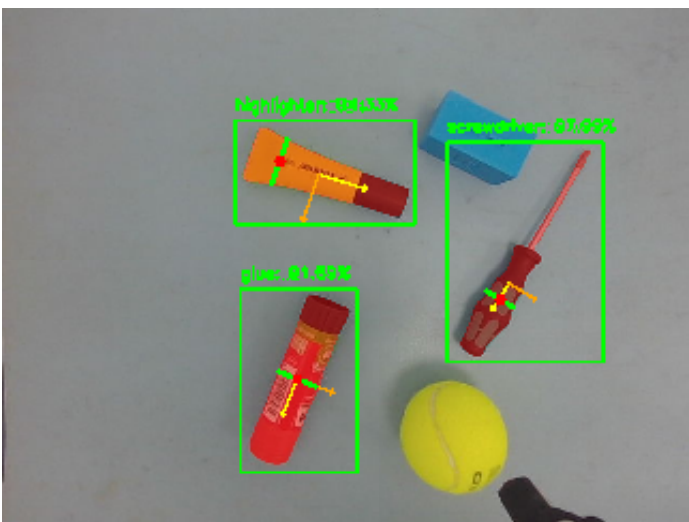


Ideal grasp point

After extracting the object to pick, it is detected. Then, the ideal pick point is computed based on the segmented mask obtaining the geometric centroid approximating the center of masses of the object, and shifting it towards a more stable and graspable point. The orientation is then computed placing the gripper parallel to the objects longest dimension.



DETECTION + SEGMENTATION



PICK POINT COMPUTATION

Results

A fully functional system has been obtained with:

- High positional picking precision.
- Reliable picking of objects of different shapes and sizes.
- Total pipeliene latency: (~4s).

Conclusions

The project successfully achieved all the objectives, building a system capable of understanding and executing voice commands to perform pick-and-place as an assistive tool for individuals with reduced mobility. It goes beyond pre-programmed instructions, enabling users to interact with their environment and manipulate objects by speaking showing the potential of combining AI with collaborative robotics.

